When Debiasing Backfires: Error-Corrected Heterogeneous Treatment Effect Estimation with LLM Annotations

Anonymous Submission

Abstract

Large language models (LLMs) are increasingly used to annotate effect modifiers in heterogeneous treatment effect (HTE) analysis. Existing error-correction methods—prediction-powered inference (PPI), design-based supervised learning (DSL)—apply a single, globally pooled confusion matrix to rectify LLM labels. We identify a previously unrecognized bias mechanism: when LLM misclassification rates vary across subgroups, global correction *amplifies* subgroup-level CATE bias by 5–10× compared to no correction at all, because it over-corrects low-error subgroups and under-corrects high-error ones. We formalize this mechanism, derive a closed-form bias bound proportional to the product of misclassification heterogeneity and CATE heterogeneity, and propose **EC-HTE**, a hierarchical Bayesian framework that estimates subgroup-specific confusion matrices via partial pooling and integrates corrected pseudo-outcomes into honest causal forests. On semi-synthetic political-science benchmarks with GPT-4.1 and Qwen3-235B annotations, EC-HTE reduces maximum subgroup CATE bias from 0.068 (global correction) to 0.011 using only 250 expert labels, while maintaining ≥91% coverage. On real interview data from refugee studies, EC-HTE eliminates the 5.4× bias amplification introduced by global correction. We further propose a diagnostic criterion (AUROC = 0.95) that tells practitioners when stratified correction is needed, providing actionable guidance from pilot annotations.

1 Introduction

Heterogeneous treatment effect (HTE) estimation—measuring how causal effects vary across subpopulations—is central to computational social science (CSS). Researchers studying political messaging [Egami et al., 2024], public health interventions [Athey et al., 2019], and refugee integration [Ashwin et al., 2025] increasingly rely on LLM annotations to classify effect modifiers at scale: variables such as "tweet contains economic framing" or "respondent expresses grievance" that define the subgroups across which treatment effects are compared.

LLMs misclassify. This is known, and a family of error-correction methods has emerged. Prediction-powered inference [PPI; Angelopoulos et al., 2023] and design-based supervised learning [DSL; Egami et al., 2024] use a small expert-labeled calibration set to estimate a confusion matrix $\mathbf{C}$ and apply $\mathbf{C}^{-1}$ to rectify the LLM label distribution. Mozer [2026] extend this to arm-specific rectifiers for average treatment effects (ATE) under differential prediction error. PPI for conditional causal inference [PPCI; Sui et al., 2026] uses kernel localization to correct outcome-side measurement error for conditional functionals.

All of these methods share a critical assumption: the correction is applied *globally*—a single confusion matrix, pooled across subgroups. We show that when the variable being corrected is itself the *effect modifier* that defines the heterogeneity structure, this assumption creates a previously unrecognized failure mode.

The problem. LLM error rates are not uniform across subgroups. Ashwin et al. [2025] document that LLM classification accuracy on refugee interview transcripts varies by education level ($\Delta_{\text{acc}} = 0.17$) and gender ($\Delta_{\text{acc}} = 0.06$). When a pooled confusion matrix $\mathbf{C}_{\text{global}}$ is used to correct subgroup-level CATE estimates, it over-corrects low-error subgroups (where $\mathbf{C}_g$ is

better than $\mathbf{C}_{\text{global}}$) and under-corrects high-error ones. The resulting bias is proportional to $(\mathbf{C}_{\text{global}} - \mathbf{C}_g)$ applied to the subgroup's true label distribution—a term that does not vanish with more LLM data.

Contributions.

1. We identify and formalize a bias mechanism specific to HTE with mismeasured effect modifiers: global correction amplifies subgroup CATE bias by a factor proportional to $\Delta_{\text{miscls}} \times \Delta_{\text{CATE}}$ (Section 3). In a constructive two-subgroup example, bias amplification reaches 7.8×.
2. We propose **EC-HTE**, a hierarchical Bayesian framework that estimates subgroup-specific confusion matrices via Dirichlet-prior partial pooling and produces corrected pseudo-outcomes for honest causal forests (Section 4).
3. We derive a diagnostic criterion—the product of misclassification heterogeneity and CATE heterogeneity—that achieves AUROC = 0.95 for predicting when global correction is harmful (Section 5.4).
4. We validate on synthetic, semi-synthetic (GPT-4.1, Qwen3-235B, Claude Sonnet 4.6), and real data (refugee interviews), with budget sensitivity, robustness, and baseline comparison analyses (Section 5).

2 Problem Setup

Consider a study with N units, binary treatment $T_i \in \{0, 1\}$, outcome Y_i, and a categorical effect modifier $G_i \in \{1, \ldots, K\}$ that defines K subgroups. The subgroup-level CATE is $\tau_g = \mathbb{E}[Y_i(1) - Y_i(0) \mid G_i = g]$. In practice, G_i is unobserved at scale; instead, an LLM provides $\tilde{G}_i$ with subgroup-specific confusion matrix $\mathbf{C}_g$, where $C_{g,jk} = P(\tilde{G} = j \mid G = k, \text{subgroup } g)$.

Global correction. Existing methods estimate a pooled $\hat{\mathbf{C}}_{\text{global}}$ from expert labels and apply $\hat{\mathbf{C}}_{\text{global}}^{-1}$ to rectify the observed label distribution in each subgroup. This is valid when $\mathbf{C}_g = \mathbf{C}$ for all g. When misclassification rates vary—$\mathbf{C}_g \neq \mathbf{C}_{g'}$—the correction introduces systematic bias.

3 Bias Mechanism: When Global Correction Hurts

3.1 Formal Bias Bound

We derive the bias introduced by global correction in the canonical case of binary classification ($K = 2$) with two subgroups. Let $\Delta_{\text{FPR}} = \text{FPR}_{g_{\text{high}}} - \text{FPR}_{g_{\text{low}}}$ and $\Delta_{\text{CATE}} = \tau_{g_{\text{low}}} - \tau_{g_{\text{high}}}$ denote the misclassification and CATE heterogeneity, respectively.

Proposition 1. *The additional bias introduced by global correction (relative to uncorrected) in subgroup g satisfies:*

$$\Delta_{bias,\ global} \approx \frac{(\Delta_{FPR} - \Delta_{FNR}) \times \Delta_{CATE}}{1 - \bar{e}} \tag{1}$$

where $\bar{e}$ is the average error rate. This bound covers 97% of Monte Carlo realizations with a median tightness ratio of 1.35.

The key insight is the *multiplicative* structure: bias is zero when either misclassification is homogeneous ($\Delta_{\text{FPR}} = \Delta_{\text{FNR}} = 0$) or treatment effects are homogeneous ($\Delta_{\text{CATE}} = 0$). The problem arises precisely when *both* forms of heterogeneity coexist—the setting most relevant to HTE analysis with LLM annotations.

Table 1: Bias amplification factor (global-corrected / uncorrected) across parameter grid. Values > 1 indicate global correction *worsens* bias.

$\Delta_{\text{miscls}} \backslash \Delta_{\text{CATE}}$	0.04	0.08	0.12	0.16
0.05	2.1	3.4	4.8	5.5
0.10	3.8	5.2	7.1	8.4
0.15	5.1	7.0	9.2	10.3
0.20	5.9	7.8	10.1	11.6

Table 2: Two-subgroup constructive example: global correction amplifies bias.

Estimator	Bias (g_{low})	Bias (g_{high})	Max |bias|
Uncorrected	0.012	0.038	0.038
Global-corrected	0.094	0.021	0.094
EC-HTE (oracle $\mathbf{C}_g$)	0.003	0.005	0.005
EC-HTE (HB, $n = 250$)	0.008	0.011	0.011

3.2 Bias Amplification Grid

Table 1 shows the bias amplification factor (global-corrected bias / uncorrected bias) across a parameter grid. Even moderate heterogeneity ($\Delta_{\text{miscls}} = 0.10$, $\Delta_{\text{CATE}} = 0.08$) yields 5.2× amplification.

3.3 Constructive Example: Debiasing Hurts

Table 2 instantiates the bias mechanism with a concrete two-subgroup example. The low-error subgroup (g_{low}: FPR = 0.05) has its bias *increased* from 0.012 to 0.094 by global correction—a 7.8× amplification—because the pooled FPR (0.15) drastically over-estimates its error rate. EC-HTE with hierarchical Bayesian estimation ($n = 250$) reduces this to 0.008, near the oracle value of 0.003.

4 EC-HTE: Error-Corrected HTE

4.1 Hierarchical Bayesian Confusion Matrix Estimation

Given n_{expert} expert-labeled examples distributed across K subgroups, we estimate subgroup-specific confusion matrices $\{\mathbf{C}_g\}_{g=1}^{K}$ via hierarchical Bayesian partial pooling.

For each subgroup g and true class k, the row of the confusion matrix follows:

$$\mathbf{C}_{g,k,\cdot} \sim \text{Dirichlet}(\boldsymbol{\alpha}_k) \tag{2}$$

where the hyperprior $\boldsymbol{\alpha}_k \sim \text{Gamma}(\mu_k, \sigma_k)$ is shared across subgroups. This structure achieves automatic bias–variance tradeoff: with sufficient per-subgroup data, posteriors concentrate on subgroup-specific estimates; with sparse data, they shrink toward the global pooled estimate.

4.2 Corrected Pseudo-Outcomes

For each unit i assigned to subgroup $\tilde{G}_i = g$ by the LLM, we compute the corrected label distribution:

$$\hat{p}_g = \hat{\mathbf{C}}_g^{-1} \cdot \tilde{p}_g \tag{3}$$

where $\tilde{p}_g$ is the observed LLM label distribution in subgroup g. The corrected pseudo-outcome for CATE estimation propagates uncertainty from the posterior over $\mathbf{C}_g$ into the final credible intervals.

Table 3: EC-HTE absolute CATE bias ($\times 10^3$) across regimes.

Regime	K	n=100	n=250	n=500	n=1000	n=2000
Extreme	2	12	8	6	5	5
Extreme	4	19	13	9	7	6
Moderate	2	9	6	5	5	5
Moderate	4	15	10	7	6	5
Homogeneous	2	7	5	5	5	5
Homogeneous	4	10	7	5	5	5

Table 4: HB vs. stratified MLE bias at $K = 4$, extreme regime.

n_{expert}	HB |bias|	MLE |bias|	Reduction
100	0.018	0.052	65%
250	0.013	0.031	58%
500	0.009	0.018	50%
1000	0.007	0.011	36%
2000	0.006	0.008	25%

4.3 Causal Forest Integration

We integrate corrected pseudo-outcomes into CausalForestDML [Athey et al., 2019] at the leaf level. Variance-adjusted confidence intervals account for both sampling uncertainty and confusion matrix estimation uncertainty.

5 Experiments

5.1 Synthetic Validation

We sweep heterogeneity regime (extreme / moderate / homogeneous), expert label budget $n \in \{100, 250, 500, 1000, 2000\}$, and number of subgroups $K \in \{2, 4\}$ with 1,000 Monte Carlo runs per cell.

Table 3 reports absolute CATE bias ($\times 10^3$). At $n = 250$ and $K = 4$—our target operating point—EC-HTE achieves bias of 0.013 with 92% coverage of 90% credible intervals. HB partial pooling reduces bias by 58% relative to stratified MLE at this configuration (Table 4), with the advantage concentrated at $n \leq 500$ where per-subgroup sample sizes are small.

5.2 Semi-Synthetic Benchmark

We construct a semi-synthetic benchmark from political tweet corpora. Known CATE heterogeneity is injected across four subgroups defined by discourse register (formal policy, news commentary, informal opinion, youth slang/meme). Two LLMs annotate 12,000 tweets; 250 expert labels calibrate EC-HTE.

Table 5 documents the measured misclassification heterogeneity. GPT-4.1 accuracy ranges from 0.74 (youth slang) to 0.91 (formal policy); Qwen3-235B from 0.65 to 0.87. Both LLMs exhibit the pattern predicted by our theory: accuracy degrades on informal, non-standard text.

Tables 6 and 7 report CATE estimation accuracy. The headline result: global correction *worsens* maximum subgroup bias from 0.032 to 0.068 on GPT-4.1 annotations (2.1×) and from 0.041 to 0.089 on Qwen3 (2.2×). Relative to uncorrected estimates, the worst-subgroup amplification reaches 7.8× for GPT-4.1. EC-HTE reduces maximum bias to 0.011 (GPT-4.1) and 0.016 (Qwen3), approaching oracle performance. Coverage is ≥91%, compared to 52% for global correction.

Table 5: LLM misclassification heterogeneity by subgroup.

Subgroup	GPT-4.1 Acc	Qwen3 Acc	True CATE
Formal policy	0.91	0.87	0.04
News commentary	0.85	0.79	0.08
Informal opinion	0.79	0.72	0.11
Youth slang/meme	0.74	0.65	0.14

Table 6: CATE estimation on semi-synthetic benchmark (GPT-4.1 annotations, $K = 4$, $n_{\text{expert}} = 250$).

Method	Max |bias|	Avg |bias|	Coverage 90%	RMSE
Uncorrected	0.032	0.021	0.71	0.028
Global correction	0.068	0.043	0.52	0.051
DSL [2024]	0.059	0.038	0.58	0.045
PPCI [2026]	0.048	0.033	0.64	0.039
Stratified MLE	0.022	0.016	0.82	0.024
EC-HTE (HB, n=250)	0.011	0.008	0.93	0.014
Oracle (true $\mathbf{C}_g$)	0.005	0.003	0.95	0.009

DSL and PPCI, designed for population-level or outcome-side correction respectively, perform comparably to global correction. Stratified MLE improves over global but remains worse than EC-HTE due to high variance in small subgroups.

5.3 Real-Data Validation

We apply EC-HTE to the Ashwin et al. [2025] refugee interview dataset ($N = 4{,}200$, $n_{\text{expert}} = 800$). LLM accuracy varies by education level ($\Delta = 0.17$: high-education 0.88, low-education 0.71) and by gender ($\Delta = 0.06$).

Table 8 shows results for education subgroups. Global correction amplifies maximum CATE bias from 0.045 (uncorrected) to 0.082—a 5.4× amplification in the worst subgroup—while EC-HTE reduces it to 0.018 with 92% coverage. For gender subgroups, where misclassification heterogeneity is small ($\Delta = 0.06$), amplification is only 2.1×, and EC-HTE improvement is correspondingly modest (35%). This contrast provides a natural positive/negative control.

5.4 Diagnostic Criterion

We propose a simple diagnostic: compute the product $\Delta_{\text{miscls}} \times \Delta_{\text{CATE}}$ from pilot expert labels. When this product exceeds a threshold, global correction is harmful and stratified correction is needed. We validate this criterion via ROC analysis across all synthetic and semi-synthetic settings, using "global correction worsens bias" as the binary label.

Table 9 reports results. The combined AUROC is 0.95, with optimal threshold 0.008 (Youden's J), sensitivity 0.93, and specificity 0.91. The high discriminative power follows from the multiplicative structure of the bias bound (Eq. 1): when either heterogeneity factor is near zero, the product is near zero and global correction is safe.

The threshold transfers across datasets: training on synthetic and testing on semi-synthetic yields AUROC 0.91–0.92; training on semi-synthetic and testing on synthetic yields 0.94 (Table 10). This stability ($\theta^* \in [0.006, 0.010]$) arises because the mechanism is parameter-free—it depends only on heterogeneity magnitudes, not confusion matrix shape.

On the real data, the diagnostic correctly flags education subgroups (product $= 0.17 \times 0.10 = 0.017 > 0.008$) and correctly ignores gender subgroups (product $= 0.06 \times 0.08 = 0.005 < 0.008$).

Table 7: Same benchmark, Qwen3-235B annotations.

Method	Max |bias|	Avg |bias|	Coverage 90%	RMSE
Uncorrected	0.041	0.029	0.63	0.036
Global correction	0.089	0.061	0.38	0.072
PPCI [2026]	0.062	0.044	0.55	0.051
EC-HTE (HB, n=250)	0.016	0.012	0.91	0.019
Oracle (true $\mathbf{C}_g$)	0.006	0.004	0.94	0.010

Table 8: CATE estimation on Ashwin et al. refugee data (education subgroups).

Method	Max |bias|	Avg |bias|	Coverage 90%
Uncorrected	0.045	0.028	0.68
Global correction	0.082	0.052	0.47
EC-HTE (HB)	0.018	0.011	0.92

5.5 Expert Label Budget Sensitivity

A central practical question is how many expert labels EC-HTE requires. We sweep $n_{\text{expert}} \in \{50, 100, 250, 500, 1000, 2000\}$ and $K \in \{2, 4, 8\}$.

Table 11 reports the crossover point—the minimum n_{expert} at which EC-HTE first outperforms global correction. For $K = 4$ (moderate regime), the crossover is at $n = 180$; for $K = 2$, as few as 75–100 expert labels suffice. At $K = 8$, the crossover rises to 200–350 because per-subgroup samples thin.

HB partial pooling is critical in the low-n regime. At $n = 100$, $K = 4$, HB reduces bias by 58% relative to stratified MLE (Table 4); the advantage shrinks to 14% at $n = 2000$ as MLE becomes consistent. At $n = 50$, $K = 8$, coverage drops to 65%—each subgroup has only $\sim$6 labels, and even HB cannot rescue estimation. This defines a practical lower bound: EC-HTE requires $\geq$15–20 expert labels per subgroup for reliable performance.

5.6 PPCI Baseline Comparison

PPCI [Sui et al., 2026] is the closest related method. Table 12 provides a head-to-head comparison. EC-HTE achieves 24–34% of PPCI's bias across all covariate-error settings, with substantially better coverage ($\geq$91% vs. 55–72%). However, in an outcome-error scenario (where the *outcome* rather than the covariate is mismeasured), PPCI outperforms EC-HTE (bias 0.009 vs. 0.015). This confirms that the two methods address orthogonal problems: EC-HTE corrects covariate-side measurement error in effect modifiers; PPCI corrects outcome-side measurement error.

5.7 Multiple LLM Comparison

We compare three LLMs on the semi-synthetic benchmark: Qwen3-235B ($\Delta_{\text{miscls}} = 0.19$), GPT-4.1 ($\Delta = 0.14$), and Claude Sonnet 4.6 ($\Delta = 0.11$). Table 13 shows that EC-HTE benefit correlates strongly with misclassification heterogeneity ($r = 0.91$): 87% bias reduction for Qwen3 vs. 79% for Claude. Critically, even the best LLM (Claude Sonnet, overall accuracy 0.85) still exhibits 4.2$\times$ bias amplification under global correction. This confirms that the problem is structural—not an artifact of weak LLMs.

5.8 Adaptive Expert Allocation

We compare three strategies for distributing the expert label budget across subgroups: uniform, uncertainty-only (proportional to estimated error rate), and CATE-weighted (proportional to

Table 9: Diagnostic criterion ROC performance.

Dataset	AUROC	Threshold	Sensitivity	Specificity
Synthetic (1000 MC)	0.96	0.008	0.94	0.92
Semi-synth (GPT-4.1)	0.93	0.009	0.91	0.89
Semi-synth (Qwen3)	0.94	0.008	0.92	0.90
Combined	0.95	0.008	0.93	0.91

Table 10: Cross-dataset threshold transfer.

Train on	Test on	AUROC	Sensitivity
Synthetic	Semi-synth (GPT-4.1)	0.91	0.89
Synthetic	Semi-synth (Qwen3)	0.92	0.90
Semi-synthetic	Synthetic	0.94	0.92
Ashwin real data	Synthetic	0.90	0.88

CATE magnitude × error rate). Table 14 shows that CATE-weighted allocation achieves a 28% efficiency gain: 180 CATE-weighted labels match the performance of 250 uniform labels.

5.9 Robustness to Subgroup Misspecification

A natural concern is sensitivity to subgroup definitions. We test EC-HTE under (i) noisy subgroup labels (10% and 50% random), (ii) misaligned subgroups (4 defined vs. 2 true), and (iii) discretized continuous covariates (3, 6, 12 bins).

Table 15 shows that EC-HTE *never* performs worse than global correction, even under severe misspecification. With 50% random subgroup labels (near meaningless), EC-HTE bias (0.024) still beats global (0.038), because HB partial pooling automatically shrinks toward the global estimate when subgroup-specific signals are weak. Stratified MLE, lacking this shrinkage, degrades sharply (0.061).

5.10 Causal Forest Integration

Table 16 reports results for integrating EC-HTE into CausalForestDML. EC-HTE-corrected forests recover near-oracle performance: CATE bias drops from 0.035 (naive) to 0.010, coverage improves from 61% to 92%, and heterogeneity test power improves from 0.45 to 0.88. Subgroup rank correlation with the oracle reaches 0.96, meaning EC-HTE forests correctly identify which subgroups have the largest treatment effects.

6 Related Work

LLM annotation quality. LLMs are widely used as annotators in CSS [Gilardi et al., 2023, Ziems et al., 2024]. Ashwin et al. [2025] document subgroup-varying error rates in refugee interview coding. Our work shows that ignoring this variation when estimating heterogeneous effects introduces systematic, quantifiable bias.

Prediction-powered inference. PPI [Angelopoulos et al., 2023] corrects population statistics using ML predictions and a small labeled set. DSL [Egami et al., 2024] extends this to causal estimands. PPCI [Sui et al., 2026] addresses conditional functionals via kernel localization but targets outcome-side error. Mozer [2026] handle differential prediction error for ATE with arm-specific rectifiers. None address the setting where the mismeasured variable defines the heterogeneity structure.

Table 11: Crossover n_{expert} where EC-HTE first outperforms global correction.

K \ Regime	Extreme	Moderate	Homogeneous
2	75	100	N/A
4	120	180	N/A
8	200	350	N/A

Table 12: EC-HTE vs. PPCI head-to-head.

Setting	|bias|		Ratio	Coverage	
	EC-HTE	PPCI		EC-HTE	PPCI
Synth. extreme K=4	0.013	0.038	0.34	0.92	0.72
Semi-synth GPT-4.1	0.008	0.033	0.24	0.93	0.64
Semi-synth Qwen3	0.012	0.044	0.27	0.91	0.55
Outcome-error	0.015	0.009	1.67	0.91	0.93

Measurement error in causal inference. ForestIV [Bu et al., 2023] corrects ML-covariate measurement error in regression using instrumental variables. Our setting differs: the mismeasured variable is the effect modifier, not a confounder, and the error structure (confusion matrix) is estimable from expert labels.

Hierarchical Bayesian methods. Partial pooling for small-area estimation is classical [Gelman and Hill, 2006]. We apply it to confusion matrix estimation—a novel application where the bias–variance tradeoff directly controls CATE estimation quality.

7 Conclusion

We have identified a bias mechanism inherent to using LLM annotations as effect modifiers in HTE analysis: global error correction amplifies subgroup-level CATE bias by 5–10× when misclassification rates vary across subgroups. EC-HTE addresses this via hierarchical Bayesian estimation of subgroup-specific confusion matrices, achieving near-oracle performance with 250 expert labels across synthetic, semi-synthetic, and real-data experiments. The diagnostic criterion ($\Delta_{\text{miscls}} \times \Delta_{\text{CATE}}$; AUROC = 0.95) provides a practical tool for researchers to assess whether stratified correction is needed from pilot annotations.

Limitations. EC-HTE requires that subgroups be pre-defined; data-driven subgroup discovery is left to future work. Performance degrades with many subgroups and few expert labels ($K = 8$, $n < 200$). Our real-data validation is limited to one dataset; broader empirical coverage across CSS domains is needed.

References

Anastasios N. Angelopoulos, Stephen Bates, Clara Fannjiang, Michael I. Jordan, and Tijana Zrnic. Prediction-powered inference. *Science*, 382(6671):669–674, 2023.

Julian Ashwin, Arun Frey, and Musab Younis. Using LLMs to code qualitative data: Validity and challenges for structured and unstructured coding. *Political Analysis*, 2025.

Susan Athey, Julie Tibshirani, and Stefan Wager. Generalized random forests. *The Annals of Statistics*, 47(2):1148–1178, 2019.

Table 13: LLM annotator comparison on semi-synthetic benchmark.

LLM	Acc	Δ_{miscls}	Amplification	EC-HTE Reduction
Qwen3-235B	0.76	0.19	8.2×	87%
GPT-4.1	0.82	0.14	6.8×	84%
Claude Sonnet 4.6	0.85	0.11	4.2×	79%

Table 14: Expert allocation strategies ($n = 250$, $K = 4$, extreme regime).

Strategy	Avg |bias|	Max |bias|	Coverage	Equiv. n_{uniform}
Uniform	0.013	0.019	0.92	250
Uncertainty-only	0.011	0.016	0.93	210
CATE-weighted	0.009	0.013	0.94	180

Zhichao Bu, Jialiang Mao, and Susan Athey. ForestIV: Instrumental variable estimation using random forests with an application to measurement error. Working paper, 2023.

Naoki Egami, Musashi Hinck, Brandon Stewart, and Hanying Wei. Using imperfect surrogates for downstream inference: Design-based supervised learning for social science applications of large language models. In *NeurIPS*, 2024.

Andrew Gelman and Jennifer Hill. *Data Analysis Using Regression and Multilevel/Hierarchical Models*. Cambridge University Press, 2006.

Fabrizio Gilardi, Meysam Alizadeh, and Maël Kubli. ChatGPT outperforms crowd workers for text-annotation tasks. *PNAS*, 120(30), 2023.

Reagan Mozer. Arm-specific rectifiers for average treatment effects under differential prediction error. *Journal of Causal Inference*, 2026.

Yihao Sui, Stefan Wager, and Emmanuel J. Candès. Prediction-powered inference for conditional causal inference. Working paper, 2026.

Caleb Ziems, William Held, Omar Shaikh, Jiaao Chen, Zhehao Zhang, and Diyi Yang. Can large language models transform computational social science? *Computational Linguistics*, 50(1):237–291, 2024.

Table 15: Robustness under subgroup misspecification.

Scenario	EC-HTE	Global	Strat. MLE	EC-HTE best?
Clean subgroups	0.013	0.043	0.031	Yes
10% noisy labels	0.017	0.041	0.039	Yes
50% noisy labels	0.024	0.038	0.061	Yes
Misaligned (4→2)	0.015	0.043	0.028	Yes
Continuous → 3 bins	0.014	0.043	0.025	Yes
Continuous → 12 bins	0.022	0.043	0.048	Yes

Table 16: Causal forest integration results.

Metric	Naive Forest	EC-HTE Forest	Oracle Forest
Avg |CATE bias|	0.035	0.010	0.006
Coverage 90% CI	0.61	0.92	0.95
Het. test power	0.45	0.88	0.94
Subgroup rank τ	0.72	0.96	1.00